

mouthparts with a noselike shape. In insects that feed on fluids, the proboscis is often long and slender, and can usually be stowed away when not in use.

PRONOTUM The upper surface of the front segment in an insect's thorax. In some insects, it is expanded to form a shield.

PROTHORAX The front segment of an insect's thorax.

PTEROSTIGMA A small, colored spot near the leading edge on some insects' wings.

PULMONATE Having lungs. Pulmonate mollusks are air-breathing species that often live on land.

PUPA (pl. PUPAE) In insects, a stage during which the larval body is broken down, and rebuilt as an adult. During the pupal stage, the insect does not feed and usually cannot move, although some pupae may wriggle if they are touched. The pupa is protected by a hard case, which is itself sometimes wrapped in silk. See also *Chrysalis*, *Cocoon*, *Larva*.

QR

QUADRUPEDAL Moving on 4 legs.

RADIAL SYMMETRY A form of symmetry in which the body is shaped like a wheel, often with the mouth at the center.

RADULA A mouthpart that many mollusks use for rasping away at food. The radula is often shaped like a belt, and armed with many microscopic tooth-like denticles.

ROSTRUM In bugs and some other insects, a set of mouthparts that looks like a beak.

RUMINANT A hoofed mammal that has a specialized digestive system, with several stomach chambers. One of these—the rumen—contains huge numbers of microorganisms that help break down the cellulose in plant cell walls. To speed this process, a ruminant usually regurgitates its food and rechews it, a process called “chewing the cud.”

RUTTING SEASON In deer, a period during the breeding season when males clash with each other for the opportunity to mate.

S

SAPROPHAGOUS Feeding on dead and decaying matter.

SCAPE The first segment in an insect's antennae, nearest its head.

SCOLEX The head of a tapeworm. The scolex is usually equipped with suckers and/or hooks that enable the worm to fasten itself to its host.

SCUTE A shieldlike plate or scale that forms a bony covering on some animals.

SCUTELLUM A triangular shield behind the pronotum in many insects.

SEBACEOUS GLAND A skin gland in mammals that normally opens near the root of a hair. Sebaceous glands produce substances that keep skin and hair in good condition.

SEDENTARY Having a lifestyle that involves relatively little movement. This term is also used to refer to animals that do not migrate. See also *Migration*.

SEQUENTIAL HERMAPHRODITE An animal with both male and female sex organs that mature at different times.

SESSILE Fastened to a solid surface, instead of being able to move from place to place. Most sessile animals are invertebrates that live in water.

SEXUAL DIMORPHISM Showing physical differences between males and females. In animals that have separate sexes, males and females always differ, but in highly dimorphic species, such as elephant seals, the 2 sexes look very different and are often quite unequal in size. Many other species, particularly among the butterflies, fishes, and birds, show differences in color between the two sexes.

SEXUAL REPRODUCTION A form of reproduction that involves fertilization of a female cell or egg, by a male cell or sperm. This is by far the most common form of reproduction in animals. It usually involves 2 parents—one of either sex—but in some species individual animals are hermaphrodite. See also *Asexual reproduction*.

SILK A protein-based fibrous material produced by spiders and some insects. Silk is liquid when it is squeezed out of a spinneret but turns into elastic fibers when stretched and exposed to air. It has a wide variety of uses. Some animals use silk to protect themselves or their eggs, to catch prey, to glide on air currents, or to lower themselves through the air.

SIMPLE EYE An eye that has a single lens. Simple eyes are found in vertebrates and cephalopod mollusks and, despite their name, are often highly complex. Rudimentary simple eyes are also found in insects. See also *Ocellus*.

SPECIES A group of similar organisms that are capable of interbreeding in the wild, and of producing fertile offspring that resemble themselves. Species are the fundamental units used in biological classification. Some species have distinct populations that vary from each other. Where the differences are significant, and the populations biologically isolated, these forms are classified as separate subspecies.

SPERMATOPHORE A packet of sperm that is transferred either directly from the male to the female, or indirectly—for example, by being left on the ground. Spermatophores are produced by a range of animals, including salamanders (Urodeles) and some insects.

SPICULE In sponges, a needlelike sliver of silica or calcium carbonate that forms part of the internal skeleton. Spicules have a wide variety of shapes.

SPINNERET The nozzlelike organs that spiders use to produce silk.

SPIRACLE In rays and some other fishes, an opening behind the eye that lets water flow into the gills. In insects, an opening on the thorax or abdomen that lets air into the tracheal system.

STEREOSCOPIC VISION see *Binocular vision*.

STERNUM The breastbone in 4-limbed vertebrates, or the thickened underside of a body segment in arthropods. In birds, the sternum has a narrow flap.

STRIDULATE Producing sound by rubbing together 2 parts of the body.

SUBCLASS A level used in classification, between a class and an order.

SUBFAMILY A level used in classification, between a family and a genus.

SUBORDER A level used in classification, between an order and a family.

SUBSPECIES see *Species*.

SUSPENSION FEEDER An animal that feeds on particles of organic matter that are suspended in water. Suspension feeders include brittlestars, some cnidarians, and many annelid worms.

SWIM BLADDER A gas-filled bladder that most bony fishes use to regulate their buoyancy. By adjusting the gas pressure inside the bladder, a fishes can become neutrally buoyant, meaning that it neither rises nor sinks in the water column.

SYMBIOSIS Any partnership between individuals of 2 different species. Animals form symbiotic partnerships with other animals, with plants, and with microorganisms.

T

TARSUS (pl. TARSI) A part of the leg. In insects, the tarsus is the equivalent of the foot, while in vertebrates, it forms the lower part of the leg or the ankle.

TEGMEN (pl. TEGMINA) The leathery forewing of some insects, such as grasshoppers, crickets, and cockroaches.

TERRESTRIAL Living wholly or mainly on the ground.

TERRITORY An area defended by an animal, or group of animals, against other members of the same species. Territories often include useful resources that help the male attract a mate.

TEST In echinoderms, a skeleton made of small calcareous plates.

THORAX The middle region of an arthropod's body. The thorax contains powerful muscles, and bears the legs and wings, if the animal has any. In 4-limbed vertebrates, the thorax is the chest.

TIBIA (pl. TIBIAE) The shin bone in 4-limbed vertebrates. In insects, the tibia forms the part of the leg immediately above the tarsus, or foot.

TISSUE A layer of cells in an animal's body. In a tissue, the cells are of the same type, and carry out the same work. See also *Organ*.

TORPOR A sleeplike state in which the body processes slow to a fraction of their normal rate. Animals usually become torpid to survive difficult conditions, such as cold or lack of food.

TRACHEA (pl. TRACHEAE) A breathing tube, known in vertebrates as the windpipe.

TRACHEAL LUNG In arthropods, a lung that connects to the tracheal system. Tracheal lungs are also found in snakes.

TRACHEAL SYSTEM A system of minute tubes that insects and arachnids use to carry oxygen into their bodies. Air enters the tubes through openings called spiracles, and then flows through the tracheae to reach individual cells. Carbon dioxide flows in the opposite direction.

TUBERCLE A hard swelling on an animal's body.

TUSK In mammals, a modified tooth that often projects outside the mouth. Tusks have a variety of uses, including defense and digging up food. In some species, only the males have them—in which case, their use is often for sexual display.

TYMBAL ORGAN A drumlike organ on a cicada's body that can vibrate to produce a piercing sound. The 2 tymbal organs are on either side of the abdomen.

TYMPANUM The external ear membrane of frogs.

U

UNDERFUR The dense fur that makes up the innermost part of a mammal's coat. Underfur is usually soft, and is a good insulator. See also *Guard hair*.

UNGULIGRADE A gait in which only the hooves touch the ground. See also *Digitigrade*, *Plantigrade*.

UTERUS In female mammals, the part of the body that houses developing young. In placental mammals, the young are connected to the wall of the uterus via a placenta.

V

VENTRAL On or near the underside.

VERTEBRATE An animal with a backbone. Vertebrates include fishes, amphibians, reptiles, birds, and mammals.

VESTIGIAL Relating to an organ that is atrophied or nonfunctional and often reduced in size.

VIVIPAROUS Reproducing by giving birth to live young. See also *Oviparous*, *Ovoviviparous*.

WX

WARM-BLOODED see *Endothermic*.

WARNING COLORATION A combination of contrasting colors that warns that an animal is dangerous. Bands of black and yellow are a typical form of warning coloration, found in stinging insects.

WATER VASCULAR SYSTEM In echinoderms, a system of fluid-filled tubes and chambers that connects with the tube feet, giving them their shape.

WEANING In mammals, the period when the mother gradually ceases to provide milk for her young.

XYLOPHAGOUS Wood-eating.

Z

ZOOID An individual animal in a colony of invertebrates. Zooids are often directly linked to each other, and may function like a single animal.

ZOOPLANKTON see *Plankton*.